ULTRON: Autonomous AI Scientist

System Architecture & Project Overview | Built by Samar (Age 16, India)

Codebase Size	82,016 lines across 219 Python files	**Initial Focus**	Discovering Novel Solid-State Battery Materials
Hardware Used	16GB Mac mini (built completely locally)	**Verification System**	Deterministic Math & Physics Code (Not LLM text)
Longest Run So Far	16+ continuous hours (stopped by API rate limits)	**Memory System**	Graph Database + Neural Network to connect papers

1. The Core Problem: Why LLMs Cannot Do Science Alone

Most 'AI for Science' projects today are simple wrappers: they connect an LLM like ChatGPT to Google search and blindly trust whatever text it writes. But LLMs frequently hallucinate fake numbers, invent impossible chemical formulas, and take mathematical shortcuts just to complete a prompt. You cannot build real-world physical science on top of hallucinations.

ULTRON fixes this by trapping the LLM inside a deterministic verification cage. The language model is used strictly to propose creative hypotheses, but it is **never allowed to grade its own work**. Every calculation and scientific claim must pass through external code and physics rules before it is accepted.

2. How the Architecture Works in 4 Simple Layers

Layer	What It Does (In Simple Terms)	Why It Matters
1. The Swarm (Singularity Engine)	Breaks down big research goals into small tasks. Specialized AI agents act as experiment designers, data analysts, and coders working in parallel.	Runs autonomously for 12 to 24 hours without needing a human to prompt it.
2. Math & Physics Gates	Every formula and output is checked by pure Python math engines. It checks whether the chemical formula is balanced, physically realistic, and thermodynamically stable.	Instantly kills hallucinations. If the math or physics is impossible, the idea is deleted on the spot.
3. The Red Team	Specialized adversarial agents whose only job is to aggressively try to prove the hypothesis wrong, find hidden flaws, and test edge cases.	Science advances by trying to break ideas, not confirm them. Only bulletproof candidates survive.
4. The World Model (Graph Database)	LLMs forget things when the context window fills up. ULTRON saves every verified finding and paper into a graph database. A neural network connects dots across distant fields.	Discovers non-obvious links across thousands of papers that standard prompts completely miss.

3. Real Execution Log (From 16-Hour Autonomous Test Run)

Here is an actual trace from ULTRON running on my 16GB Mac mini, showing how it generates an idea, verifies it, survives attacks, and recovers from network drops:

```
[14:10:02] CENTRAL_ORCHESTRATOR: Starting 12-hour research loop on solid-state battery electrolytes.
[15:22:18] HYPOTHESIS_AGENT: Proposed new lithium-sulfur electrolyte composition candidate #142.
[15:22:20] MATH_VERIFICATION_GATE: Checking atomic ratios, charge balance, and chemical valencies... PASSED.
[15:25:40] STABILITY_CHECK: Testing stability against decomposition into competing materials... PASSED (Stable).
[15:31:12] RED_TEAM_ATTACK: Testing temperature limits and ionic conductivity claims... PASSED (Survived attack).
[15:35:00] WORLD_MODEL: Candidate #142 verified and permanently stored in graph database.
[18:42:15] SYSTEM_MONITOR: External API rate limit reached (free tier limit hit).
[18:42:16] SELF_HEALING_ENGINE: Full system state and memory saved to disk checkpoint. Zero progress lost.
```

4. Roadmap: What the ■3.5L–■4.2L Grant Actually Unlocks

The software engine is 100% built. What is holding it back right now is hardware: running local databases and agents on a 16GB Mac mini chokes RAM, and free-tier APIs shut down long runs. The grant will be used 100% for infrastructure to run 50 continuous discovery campaigns:

Timeline	What Will Be Built & Shipped	Where the Money Goes
Month 1	Migrate ULTRON from local Mac mini to dedicated Cloud GPUs (RunPod/Lambda); connect verified open crystal databases.	■1,20,000 (Cloud GPU Servers)
Months 2–3	Run 50 continuous autonomous discovery campaigns for novel solid-state battery electrolytes without rate-limit stops.	■1,80,000 (Frontier API Tokens)
Month 4	Publish all verified candidate materials, code, and simulation reports openly to the scientific community.	■50,000 (Storage & Datasets)

Note: Zero rupees go to personal salary, food, or office space. I work from my room. Every single rupee is used as fuel to run autonomous scientific campaigns.